

Yeeun Shin

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RESEARCH INTERESTS

Human-Computer Interaction, Extended Reality (XR), Adaptive AI Interface, Creativity Support
My research explores tangible and multimodal XR interfaces that harness embodied cognition to amplify human creativity. I investigate interaction techniques for adaptive XR+AI systems that transform instinctive movement into expressive and participatory workflows.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2021 – Jun 2023
M.S. in Industrial Design (Specialization: Human-Computer Interaction) Daejeon, Korea

- Thesis: *Immersive Authoring Interface with 3D Virtual Controls on Physical Desk*
- Thesis Committee: Woohun Lee, Seok-Hyung Bae, Andrea Bianchi
- Advisor: Prof. Woohun Lee

Pohang University of Science and Technology (POSTECH) Mar. 2016 – Feb. 2021
B.S. in Materials Science and Engineering Pohang, Korea

- Graduated *Cum Laude*

Institut National des Sciences Appliquées de Lyon (INSA Lyon) Aug. 2019 - Jan. 2020
Exchange Student, Materials Science and Engineering Villeurbanne, France

PUBLICATIONS

Peer-reviewed publications in top-tier venues for HCI and interactive techniques. (equal contribution)*

[5] Desk Console: Augmenting 3D Virtual Controls on Physical Desks for Immersive Authoring
Yeeun Shin, Seung Hyeon Han, Woohun Lee
In *ACM CHI Conference on Human Factors in Computing Systems (CHI EA'25)*

🏆 Student Game Competition Finalist

[4] Spatial Chef: A Spatial Transforming VR Game with Full Body Interaction
Yeeun Shin*, Yewon Lee*, Sungbaek Kim*, Soomin Park*
In *ACM CHI Conference on Human Factors in Computing Systems (CHI EA'23)*

🏆 Best in Show Honorable Mention

[3] WonderScope: Practical Near-surface AR Device for Museum Exhibits
HyeonBeom Yi, **Yeeun Shin**, Sehee Lee, Eunhye Youn, Auejin Ham, Geehyuk Lee, Woohun Lee
In *ACM SIGGRAPH 2022 Emerging Technologies*

[2] ProjecString: Turning an Everyday String Curtain Into an Interactive Projection Display
Wooje Chang*, **Yeeun Shin***, Yeon Soo Kim*, Woohun Lee
In *ACM SIGGRAPH 2022 Posters*

[1] ChromoFilament: Designing a Thermochromic Filament for Displaying Malleable States
Donghyeon Ko, **Yeeun Shin**, Junbeom Shin, Jiwoo Hong, Woohun Lee
In *ACM Designing Interactive Systems Conference (DIS '22)*

AWARDS & HONORS

- Student Game Competition Finalist | ACM CHI 2023
- Emerging Technologies Honorable Mention (Top 3) | ACM SIGGRAPH 2022
- iF Design Award, Winner – UX 2023
- 1st Place, AI Idea Competition | LG CNS 2018
- National Merit Scholarship for Science and Engineering | Korea Student Aid Foundation 2018
- Highest Academic Achievement Scholarship | POSTECH 2018
- Academic Excellence Scholarship | POSTECH 2018

RESEARCH EXPERIENCE	Research Assistant WonderLab, KAIST Advised by Prof. Woohun Lee	Mar. 2021 – Aug 2023 Daejeon, Korea
	▪ Tangible XR Interfaces to Support Embodied Creativity Led design of XR authoring interface augmenting virtual panels as spatial controls on physical desks; identified workflow gaps via contextual inquiry and evaluated user behaviors; demoed at CHI '25 [5]. ▪ Interactive Materials for Creative Fabrication Developed thermochromic filament that visualizes malleable states to support creative decisions during fabrication; derived color mappings in design workshops and evaluated effects in user studies [1]. ▪ Multimodal Interaction for XR Systems • <i>NRF-funded</i>: Developed multimodal near-surface AR applications using sensor-fusion tracking data responsive to user movement; deployed interactive content in museums [3]. • <i>Samsung-funded</i>: Designed inclusive XR interactions via participatory design and expert interviews; developed gesture-based interfaces for AR glasses using Figma and Unity.	
	Visiting Researcher HCI Lab, KAIST Advised jointly by Prof. Woohun Lee and Prof. Geehyuk Lee	Jan. 2022 – Apr 2022 Daejeon, Korea
PROFESSIONAL EXPERIENCE	• Led interaction design and derived micro-gesture heuristics for vision-based gestural input systems.	
	AI Interaction Designer Samsung Electronics	Jan. 2024 – Present Seoul, Korea
	• Spearheaded Gemini-integrated AI agent for Smart TVs through user studies and cross-functional collaboration with Google Cloud; inventor on 8 AI interaction patents. • Conducted human-AI interaction research, prototyping web app visualizing AI reasoning and enabling participatory decision-making with human-like agents.	
	UX Intern MXXR	Nov. 2020 – Mar. 2021 Seoul, Korea
	• Led onboarding UX flow for mobile AR app with camera-based spatial sensing.	
	Software Engineering Intern LG CNS Research Center	Jun. 2018 – Aug. 2018 Seoul, Korea
	• Built an Android smartwatch app for real-time factory task tracking, integrating Bluetooth beacons and context-aware UI for industrial IoT environments.	
	Interactive Prototyping Intern Geekble	Jan. 2018 - Feb. 2018 Seoul, Korea
	• Developed sensor-integrated Arduino prototypes embedded in everyday objects detecting user motion for context-aware interaction.	
ACADEMIC ACTIVITIES	EXHIBITION & PRESENTATION	
	• Presenter, CHI Interactivity Demo • Presenter, CHI Student Game Competition • Presenter, SIGGRAPH Emerging Technologies Demo • Selected Poster Presenter, SIGGRAPH Art Papers Roundtable • Research Featured on KBS, MBC, TJB (National Broadcasting) • Research Exhibitor, Korea National Science Museum Special Exhibition • Research Exhibitor, Korea National Science Museum Living Lab • Research Exhibitor, Gwacheon National Science Museum	Japan 2025 Germany 2023 Canada 2022 Canada 2022 2022 2022 2021 2021
	SERVICE	
	• Reviewer (CHI 2026), Student Volunteer (TEI 2022) • Teaching Assistant, Design Entrepreneurship (KAIST ID402)	
SKILLS	Programming	Unity3D (C#), JavaScript/TypeScript, Python, C, C++, Java, HTML, CSS, Git
	Prototyping	Oculus SDK, XR Interaction Toolkit, Arduino, Raspberry Pi, Processing, 3D Printing, Laser Cutting, CNC, Rhino/Grasshopper, Figma, Sketch, Adobe CC
	Research	(qualitative) Focus Group, Contextual Inquiry, User study design, Thematic analysis (quantitative) Statistical & Data Analysis (SPSS, Python, SQL)